



First Semester MBA Degree Examination, Dec.2024/Jan.2025
Business Statistics

Max. Marks: 100

Note: 1. Answer any FOUR full questions from Q.No.1 to 7.
2. Q.No. 8 is compulsory.
3. M : Marks, L: Bloom's level, C: Course outcomes.

		M	L	C
Q.1	a. Define Statistics.	3	L1	CO1
	b. Mean and standard deviations of two distribution of 100 and 150 items were 50, 5 and 40, 6 respectively, find the mean and standard deviation of all the 150 items taken together.	7	L2	CO1
	c. Prices of a particular commodity in 5 years in two cities are given below: Price in city A : 20 22 19 23 10 Price in city B : 10 20 18 12 15 Find which city has more stable prices.	10	L3	CO2
Q.2	a. Define types of correlation with an example.	3	L1	CO1
	b. The measure of skewness for a certain distribution is -0.8. If the lower and upper quartiles are 44.1 and 56.6 respectively, find the median.	7	L2	CO1
	c. Find out Karl Pearson's co-efficient of correlation from the following data of marks obtained by 10 students in a class test. Marks in economics: 45 70 65 30 90 40 50 75 85 60 Marks in accountancy : 35 90 70 40 95 40 60 80 80 50	10	L3	CO2
Q.3	a. Explain the significance of measuring dispersion.	3	L1	CO1
	b. Discuss difference between parametric and non-parametric test.	7	L1	CO1
	c. Ten competitors in a chess tournament are ranked by three judges in the following order: Judge 1 - 1 5 4 8 9 6 10 7 3 2 Judge 2 - 4 8 7 6 5 9 10 3 2 1 Judge 3 - 6 7 8 1 5 10 9 2 3 4 Use the rank correlation coefficient to discuss which pair of judges have the nearest approach.	10	L3	CO2

Q.4	a. Define mode and give 2 suitable examples.	3	L1	CO1																														
	b. Define the following terms: i) Independent events ii) Mutually exclusive events iii) Equally likely event.	7	L1	CO1																														
	c. Calculate seasonal indices for the rainfall (in mm) in Karnataka given by simple average method. <table><tr><td>Years</td><td>I</td><td>II</td><td>III</td><td>IV</td></tr><tr><td>2017</td><td>118.4</td><td>260</td><td>379.4</td><td>70</td></tr><tr><td>2018</td><td>85.8</td><td>185.4</td><td>407.1</td><td>8.7</td></tr><tr><td>2019</td><td>129.8</td><td>336.5</td><td>403.1</td><td>12</td></tr><tr><td>2020</td><td>283.4</td><td>360.7</td><td>472.1</td><td>14.3</td></tr><tr><td>2021</td><td>849.1</td><td>308.5</td><td>828.8</td><td>15.9</td></tr></table>	Years	I	II	III	IV	2017	118.4	260	379.4	70	2018	85.8	185.4	407.1	8.7	2019	129.8	336.5	403.1	12	2020	283.4	360.7	472.1	14.3	2021	849.1	308.5	828.8	15.9	10	L4	CO3
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Q.5	a. Define Hypothesis.	3	L1	CO1																														
	b. A factory has two machines, machine I produces 30% of the items of output and machine II produces 70% of the items. Further 5% of the items produced by the machine I were defective and only 1% produced by machine II were defective. If the defective item is drawn at random, what is the probability that it was produced by machine I?	7	L2	CO3																														
	c. Random samples drawn from normal population are: <table><tr><td>Sample 1:</td><td>20</td><td>16</td><td>26</td><td>27</td><td>23</td><td>22</td><td>18</td><td>24</td><td>25</td><td>19</td><td></td><td></td></tr><tr><td>Sample 2:</td><td>27</td><td>33</td><td>42</td><td>35</td><td>32</td><td>34</td><td>38</td><td>28</td><td>41</td><td>43</td><td>30</td><td>37</td></tr></table> Obtain estimate of variance of 2 population and test whether 2 populations are same.	Sample 1:	20	16	26	27	23	22	18	24	25	19			Sample 2:	27	33	42	35	32	34	38	28	41	43	30	37	10	L3	CO2				
Sample 1:	20	16	26	27	23	22	18	24	25	19																								
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Q.6	a. Define normal distribution.	3	L1	CO1																														
	b. An intelligent quotient of 16 students from one area of a city showed a mean of 107 and SD of 10. While the IQ of 14 students from another area of the city showed mean of 112 and SD of 8. Is there a significance difference between the IQ's of the 2 groups at 0.01 level of significance?	7	L2	CO1																														
	c. Explain steps in formulation of hypothesis.	10	L2	CO1																														
Q.7	a. Define time series analysis.	3	L1	CO1																														
	b. Below are given figures of production (in 000's tons) of a sugar factory. <table><tr><td>Year :</td><td>1999</td><td>2000</td><td>2001</td><td>2002</td><td>2003</td><td>2004</td><td>2005</td></tr><tr><td>Production :</td><td>77</td><td>88</td><td>94</td><td>85</td><td>91</td><td>98</td><td>90</td></tr></table> i) Fit a straight line by the least square and show the trend values. ii) What is the monthly increase in production?	Year :	1999	2000	2001	2002	2003	2004	2005	Production :	77	88	94	85	91	98	90	7	L3	CO2														
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- c. On an average a printer makes 4 printing mistakes while printing one page. What is the probability that a randomly observed page is free from mistakes? Among 300 pages, how many pages would you expect mistakes [$e^{-4} = 0.0183$].

10 L1 CO2

Q.8

CASE STUDY (Compulsory)

- a. Construct 5 year moving average of number of students studying in the college, they are:
- | | | | | | | | | | | |
|-------------------|------|------|------|------|------|------|------|------|------|------|
| Years : | 1996 | 1997 | 1998 | 1999 | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 |
| No. of students : | 332 | 317 | 357 | 392 | 402 | 405 | 410 | 427 | 405 | 431 |

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10 L1 CO2

- b. Solve the problem using 3years moving average method.

Years :	1999	2000	2001	2002	2003	2004	2005
Duration :	-3	-2	-1	0	1	2	3

10 L3 CO2
